

MATH1560J Final Exam Practice Set

Problem 1. Given the parametric curve $(x(t), y(t))$ where:

$$x(t) = 2 \cos(3t) - 3 \sin(2t), \quad y(t) = 4 \sin(3t) + 3 \cos(4t)$$

Find the equations (in terms of x, y only) for tangent lines at $t = 0$ and $t = \pi$.

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{12 \cos 3t - 12 \sin 4t}{-6 \sin 3t - 6 \cos 2t} = \frac{2 \sin 4t - 2 \cos 3t}{\sin 3t + \cos 2t}$$

① $t=0$. $x(0) = 2$. $y(0) = 3$ $y - 3 = -2(x - 2)$

$$\frac{dy}{dx} = \frac{-2}{1} = -2$$

② $t=\pi$. $x(\pi) = -2$. $y(\pi) = 3$

$$\frac{dy}{dx} = \frac{2}{1} = 2$$
 $y - 3 = 2(x + 2)$

Problem 2. Let $R \subseteq \mathbb{R}^2$ be the region bounded by the graphs of functions $y = f(x) = x^2$ and $y = g(x) = 4 - 2x - x^2$.

- (i) Write down the integral for calculating the **perimeter** (total length of boundary) of R . Then evaluate the integral.

- (ii) Let E be the solid obtained by rotating the region R around the x -axis. Write down the integral for calculating the **surface area** of E . Then evaluate the integral.

(Remark: Briefly explain your formula if you consider it too exotic for others to understand.)

Problem 2. Let $R \subseteq \mathbb{R}^2$ be the region bounded by the graphs of functions $y = f(x) = x^2$ and $y = g(x) = 4 - 2x - x^2$.

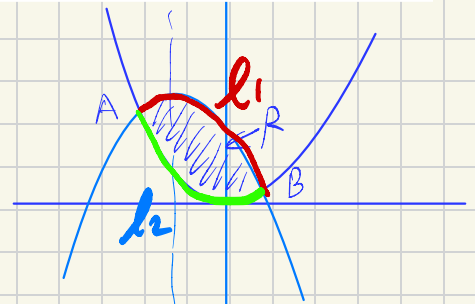
- (i) Write down the integral for calculating the **perimeter** (total length of boundary) of R . Then evaluate the integral.

① find A & B.

$$f(x) = g(x) \Rightarrow x^2 = 4 - 2x - x^2$$

$$2x^2 + 2x - 4 = 0 \quad x = -2 / 1$$

$$A(-2, 4) \quad B(1, 1)$$



② $l_1 = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{-2}^1 \sqrt{1 + (-2x-1)^2} dx = \int_{-2}^1 \sqrt{1 + (x+1)^2} dx$

$$l_2 = \int_a^b \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \int_{-2}^1 \sqrt{1 + 4x^2} dx$$

$$\int \sqrt{1 + 4x^2} dx \quad \underline{X = \frac{1}{2} \tan \theta} \quad \frac{1}{2} \int \sqrt{1 + \tan^2 \theta} d \tan \theta = \frac{1}{2} \int \sec^2 \theta d\theta = \frac{1}{2} \int \sec \theta \tan \theta d\theta = \frac{1}{2} (\sec \theta - \int \tan \theta d\theta)$$

$$\int \tan \theta d\sec \theta = \int \tan^2 \theta \sec \theta d\theta = \int \sec \theta (\sec^2 \theta - 1) d\theta = \int \sec^3 \theta d\theta - \int \sec \theta d\theta$$

thus $\int \sec^3 \theta d\theta = \sec \theta \tan \theta - \int \sec^3 \theta d\theta + \int \sec \theta d\theta$

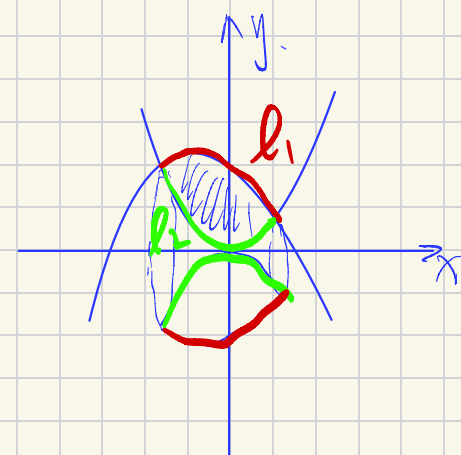
$$\int \sec^3 \theta d\theta = \frac{1}{2} \sec \theta \tan \theta + \ln |\sec \theta + \tan \theta|$$

$$P = 2\sqrt{5} + \sqrt{5} + \frac{1}{2} \ln(4 + \sqrt{5}) + \frac{1}{2} \ln(2 + \sqrt{5})$$

(ii) Let E be the solid obtained by rotating the region R around the x -axis. Write down the integral for calculating the **surface area** of E . Then evaluate the integral.

(Remark: Briefly explain your formula if you consider it too exotic for others to understand.)

① L_1 rotation $S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$



$$2\pi \int_{-1}^2 x^2 \sqrt{1 + 4x^2} dx$$

$$\int x^2 \sqrt{1 + 4x^2} dx \stackrel{x = \frac{1}{2} \tan \theta}{=} \int \frac{1}{4} \tan^2 \theta \sec \theta d\left(\frac{1}{2} \tan \theta\right)$$

$$= \frac{1}{8} \int \tan^2 \theta \sec \theta \cdot \sec^2 \theta d\theta$$

$$= \frac{1}{8} \int (\sec^2 \theta - 1) \sec^3 \theta d\theta$$

$$= \frac{1}{8} \int \sec^5 \theta - \sec^3 \theta d\theta$$

$$\int \sec^5 \theta d\theta = \int \sec^3 \theta \sec^2 \theta d\theta$$

$$= \int \sec^3 \theta d \tan \theta$$

$$= \sec^3 \theta \tan \theta - 3 \int \tan \theta \sec^3 \theta \sec \theta \tan \theta d\theta$$

$$= \sec^3 \theta \tan \theta - 3 \int (\sec^2 \theta - 1) \sec^3 \theta d\theta$$

$$\pi \left[\frac{95}{16} + \frac{33\sqrt{17}}{8} - \frac{1}{32} \ln((2+\sqrt{5})(4+\sqrt{17})) \right]$$

② L_2 rotation

$$S = 2\pi \int_a^b y \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

$$= 2\pi \int_0^b (4 - 2x - x^2) \sqrt{1 + 4(x+1)^2} dx$$

$$= 2\pi \int_0^b (5 - (x+1)^2) \sqrt{1 + 4(x+1)^2} d(x+1)$$

$$= 2\pi \int_{1+1}^{b+1} 5 \sqrt{1 + 4x^2} dx - 2\pi \int_{1+1}^{b+1} 5x^2 \sqrt{1 + 4x^2} dx$$

(1)

Problem 3. Given the function:

$$f(x) = \sum_{k \geq 0} \left(\frac{3}{5^k} + (-1)^k \frac{5}{3^k} \right) x^k$$

Find the Taylor series for $f(x)$ centered at $x = 1$. Show your work.

$$f(x) = \sum_{k \geq 0} 3 \cdot \left(\frac{x}{5}\right)^k + \sum_{k \geq 0} 5 \cdot \left(\frac{-x}{3}\right)^k$$

$$\sum_{k \geq 0} r^k = \frac{1}{1-r} \quad \uparrow$$

$$f(x) = 3 \frac{1}{1 - \frac{x}{5}} + 5 \frac{1}{1 + \frac{x}{3}} = \frac{15}{5-x} + \frac{15}{x+3}$$

因为是我找 $x=1$ 时的展开. 所以用 $t=x-1$ 替换掉. 找 $t=0$ 时的展开

$$f(t) = \frac{15}{4-t} + \frac{15}{4+t} = \frac{15 \times 8}{16-t^2}$$

$$\text{继续用 } \sum_{k \geq 0} r^k = \frac{1}{1-r}$$

$$f(t) = \frac{120}{16} \cdot \frac{1}{1 - \frac{t^2}{16}} = \frac{15}{2} \sum_{k \geq 0} \left(\frac{t^2}{16}\right)^k = \frac{15}{2} \sum_{k \geq 0} \frac{(x-1)^{2k}}{16^k}$$

Problem 4. Given $f(x) = \frac{x}{\log(1-x)}$ for $0 < x < 1$.

Evaluate the following series for $x \in (0, 1)$:

$$\sum_{n \geq 1} \frac{x^n (1-x)^n}{n!} f^{(n)}(x)$$

(Your result should be a formula in terms of x and *does not contain infinite summation*.)



$$\begin{aligned} f(x) &= f(a) + \frac{f'(a)}{1!} (x-a) + \frac{f''(a)}{2!} (x-a)^2 + \dots \\ f(x+h) &= f(x) + \frac{f'(x)}{1!} h + \frac{f''(x)}{2!} h^2 + \dots + \frac{f^{(n)}(x)}{n!} h^n \\ &= f(x) + \sum_{n \geq 1} \frac{f^{(n)}(x)}{n!} h^n \end{aligned}$$

where $h = x(1-x)$

$$f(x + (1-x) \cdot x) - f(x) = \sum_{n \geq 1} \frac{x^n (1-x)^n}{n!} f^{(n)}(x)$$

where $f(x) = \frac{x}{\log(1-x)}$

$$f(x + x - x^2) = \frac{2x - x^2}{\log(1 - 2x + x^2)} = \frac{2x - x^2}{2 \log(1-x)}$$

$$f(x) = \frac{x}{\log(1-x)}$$

$$f(x + x - x^2) - f(x) = - \frac{x^2}{2 \log(1-x)}$$

Problem 5. Find the area that is both **inside** the polar curve $r = 1 + \cos \theta$ (cardioid) and **outside** the polar curve $r = \sqrt{3} \sin \theta$ (circle).

Note: Write down the integral for calculating the area first, then evaluate the integral.

(Remark: Briefly explain your formula if you consider it too exotic for others to understand.)

l1: 指点

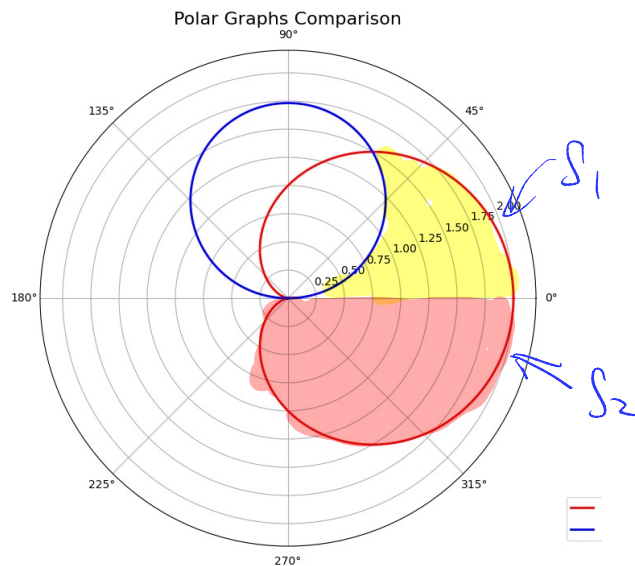
l2: $r^2 = \sqrt{3} r \sin \theta$

$$x^2 + y^2 = \sqrt{3} y$$

$$x^2 + \left(y - \frac{\sqrt{3}}{2}\right)^2 = \frac{3}{4}$$

$$S_1 = \frac{1}{2} \int_0^{\frac{\pi}{3}} (1 + \cos \theta)^2 - (\sqrt{3} \sin \theta)^2 d\theta$$

$$S_2 = \frac{1}{2} \int_{-\pi}^0 (1 + \cos \theta)^2 d\theta$$



Problem 6. Given the power series:

$$\sum_{n \geq 1} \frac{2^n (x-10)^n}{n^{1/3}}$$

- (i) Find the radius of convergence (ROC) for the series. Show your work and identify convergence tests you applied.

$$\text{for } a_n = \frac{2^n (x-10)^n}{n^{1/3}}$$

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{n+1} (x-10)^{n+1} n^{1/3}}{2^n (x-10)^n (n+1)^{1/3}} \right| = 2|x-10|$$

$$L < 1 \quad \text{ROC} = \frac{1}{2}$$

- (ii) Find the interval of convergence (IOC) for the series. Show your work and identify convergence tests you applied.

$$x-10 = 0 \rightarrow x=10.$$

$$(9.5, 10.5)$$

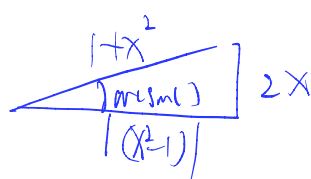
$$\textcircled{1} 9.5: \sum \frac{2^n}{n^{1/3}} \quad \text{CD} \quad \checkmark$$

$$\textcircled{2} 10.5: \sum \frac{1}{n^{1/3}} \quad \frac{1}{3} \leq 1 \quad (\text{X})$$

$$[9.5, 10.5)$$

Problem 7. Evaluate the following definite integral:

$$\int_0^2 \arcsin\left(\frac{2x}{1+x^2}\right) dx$$

$\star$ $x = \tan \frac{\theta}{2}$ 

$$\frac{2x}{1+x^2} = \frac{2 \tan \frac{\theta}{2}}{\sec^2 \frac{\theta}{2}} = \frac{2 \sin \frac{\theta}{2} \cos \frac{\theta}{2}}{\cos^2 \frac{\theta}{2}} = \sin \theta.$$

$$\arcsin\left(\frac{2x}{1+x^2}\right) = \arcsin(\sin \theta)$$

$\star$ 分类讨论 θ 范围

$x^2 - 1 \geq 0 \rightarrow x \geq 1$. Where $\tan \frac{\theta}{2} \geq 1$, $\frac{\theta}{2} \geq \frac{\pi}{4}$, $\theta \geq \frac{\pi}{2}$.

$$\arcsin(\sin \theta) = \pi - \theta.$$

$x^2 - 1 < 0 \rightarrow x < 1$ where $\tan \frac{\theta}{2} < 1$, $\theta < \frac{\pi}{2}$.

$$\arcsin(\sin \theta) = \theta$$

$$\begin{aligned} \text{thus } I &= \int_0^1 \arcsin \frac{2x}{1+x^2} dx + \int_1^2 \arcsin \frac{2x}{1+x^2} dx \\ &= \int_0^{\frac{\pi}{2}} \theta d \tan \frac{\theta}{2} + \int_{\frac{\pi}{2}}^{\pi} (\pi - \theta) d \tan \frac{\theta}{2} \end{aligned}$$

$$= \int_0^1 \arctan 2x dx + \int_1^2 (\pi - \arctan 2x) dx.$$

$$\begin{aligned} \int_0^1 \arctan 2x dx &= \frac{1}{2} \int_0^2 \arctan x dx = \frac{1}{2} \left[x \arctan x - \int x d \arctan x \right]_0^2 = \frac{1}{2} \left(x \arctan x - \frac{1}{2} \int \frac{1}{1+x^2} dx^2 \right) \Big|_0^2 \\ &= \frac{1}{2} \left(x \arctan x - \frac{1}{2} \ln(x^2+1) \right) \Big|_0^2 \end{aligned}$$

$$= 2\pi - 4 \arctan 2 + \ln 5$$

Problem 8. Find the antiderivative of f , where:

$$f(x) = \frac{x^4 + 4x^3 + 11x^2 + 12x + 8}{(x^2 + 2x + 3)^2(x + 1)}$$

$$f(x) = \frac{Ax^3 + Bx^2 + Cx + D}{(x^2 + 2x + 3)^2} + \frac{E}{x+1}$$

$$\begin{aligned} &\uparrow \\ &x^4 + 4x^3 + 9 + 4x^3 + 12x + 6x^2 \\ &= x^4 + 4x^3 + 10x^2 + 12x + 9 \end{aligned}$$

$$\begin{aligned} \text{thus } f(x) &= \frac{(x^2 + 2x + 3)^2 + x^2 - 1}{(x^2 + 2x + 3)^2(x + 1)} \\ &= \frac{x-1}{(x^2 + 2x + 3)^2} + \frac{1}{x+1} \end{aligned}$$

$$\int f(x) dx = \int \frac{1}{x+1} d(x+1) + \int \frac{x-1}{(x^2 + 2x + 3)^2} dx$$

$$= \ln(x+1) + \int \frac{x+1-2}{(x^2 + 2x + 3)^2} d(x+1)$$

$$\begin{aligned} &\stackrel{t=x+1}{=} \ln(x+1) + \int \frac{t-2}{(t^2 + 2)^2} dt = \int \frac{t}{(t^2 + 2)^2} dt - 2 \int \frac{1}{(t^2 + 2)^2} dt \end{aligned}$$

$$\stackrel{t=\sqrt{2} \tan \theta}{=} \frac{1}{2} \int \frac{1}{(t^2 + 2)^2} d(t^2 + 2) - 2 \int \frac{1}{4 \sec^4 \theta} \cdot \sqrt{2} \sec^2 \theta d\theta$$

$$= \frac{1}{2} \frac{-1}{t^2 + 2} - \frac{\sqrt{2}}{2} \int \cos^2 \theta d\theta$$

$$= -\frac{1}{2(t^2 + 2)} - \frac{\sqrt{2}}{2} \int \frac{\cos 2\theta + 1}{2} d\theta$$

$$= -\frac{1}{2(t^2 + 2)} - \frac{\sqrt{2}}{4} \left(\theta + \frac{1}{2} \sin 2\theta \right)$$

$$= -\frac{1}{2(t^2 + 2)} - \frac{\sqrt{2}}{4} \left(\arctan \frac{t}{\sqrt{2}} + \frac{2\sqrt{2}t}{2(t^2 + 2)} \right)$$

$$\begin{aligned} \int f(x) dx &= \ln(x+1) - \frac{1}{2((x+1)^2 + 2)} - \frac{\sqrt{2}}{4} \arctan \left(\frac{x+1}{\sqrt{2}} \right) - \frac{x+1}{2((x+1)^2 + 2)} + C \\ &\quad \xrightarrow{x+2} \end{aligned}$$

Appendix: Useful Power Series Expansions

- $\frac{1}{1-x} = \sum_{n \geq 0} x^n, \quad \text{ROC} = 1$
- $e^x = \sum_{n \geq 0} \frac{x^n}{n!}, \quad \text{ROC} = \infty$
- $\sin x = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{(2n+1)!}, \quad \text{ROC} = \infty$
- $\cos x = \sum_{n \geq 0} (-1)^n \frac{x^{2n}}{(2n)!}, \quad \text{ROC} = \infty$
- $\arctan x = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{2n+1}, \quad \text{ROC} = 1$
- $\log(1+x) = \sum_{n \geq 1} (-1)^{n-1} \frac{x^n}{n}, \quad \text{ROC} = 1$
- $(1+x)^\alpha = \sum_{n \geq 0} \binom{\alpha}{n} x^n, \quad \text{ROC} = 1$